

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION**

ORDER NO. 93-236

**WASTE DISCHARGE REQUIREMENTS
FOR
SPANISH FLAT WATER DISTRICT
MONTICELLO CEMETERY DISTRICT
AND NAPA COUNTY
SPANISH FLAT WASTEWATER TREATMENT AND DISPOSAL FACILITY
NAPA COUNTY**

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Spanish Flat Water District submitted a Report of Waste Discharge, dated 21 December 1992, for a wastewater treatment facility. The treatment plant and wastewater collection system are owned by Spanish Flat Water District.
2. The Board, on 31 May 1985, adopted Order No. 85-134 which prescribed requirements for a discharge from Spanish Flat Water District to evaporation/percolation ponds and a 2.5 acre spray irrigation field.
3. The treatment facility and storage pond are on property which is leased to the Spanish Flat Water District by the property owner, Napa County.
4. The Discharger will use treated wastewater to spray irrigate landscape at two areas. One irrigation area is the existing 2.5 acre spray irrigation field owned by the Spanish Flat Water District. The second area is the Monticello Cemetery which is approximately 3.7 acres in size. Monticello Cemetery is owned by the Monticello Cemetery District. Spanish Flat Water District, Monticello Cemetery District, and Napa County are jointly referred to as the Discharger.
5. The Discharger proposes to replace its existing evaporation/percolation pond system with an extended aeration treatment plant. The plant will provide sewerage service to Spanish Flat Mobile Villa, Spanish Flat Woodlands and Spanish Flat Center.
6. The wastewater treatment facility and disposal areas are in Sections 17, 18, and 20, T8N, R3W, MDB&M, with surface water drainage to Lake Berryessa, as shown in Attachment A, which is attached hereto and part of the order by reference.

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7. The treatment system is an extended aeration plant with a design average daily flow rate of 25,000 gpd, and a maximum daily flow rate of 53,000 gpd. The facility process includes an aeration tank, clarifier and chlorine contact chamber. Treated effluent will be discharged to an unlined 13 acre-foot pond.
8. The characteristics of the treated wastewater as described in the Report of Waste Discharge are:

<u>Constituents</u>	<u>Concentrations</u>
BOD ₅	30 mg/l
Total Suspended Solids	30 mg/l
Total Coliform	23 MPN/100 ml
pH	6.5 to 8.5

9. The Board adopted a Water Quality Control Plan, Second Edition, for the Sacramento River (5A) Basin (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
10. The beneficial uses of Lake Berryessa are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; ground water recharge; fresh water replenishment; hydroelectric power generation; and preservation and enhancement of fish, wildlife and other aquatic resources.
11. The beneficial uses of the underlying ground water are municipal, industrial, and agricultural supply.
12. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations, Section 60301, et seq. (hereafter Title 22) for the use of reclaimed water and has developed guidelines for specific uses. These uses are consistent with those guidelines.
13. Spanish Flat Water District has certified a final Negative Declaration in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The project as approved may have the following significant impacts on air and water quality:

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- a. The continued and expanded disposal of treated wastewater through spray irrigation may subject the community to aerosols.
 - b. Potential sources of odors in wastewater collection, treatment and disposal facilities.
 - c. Potential for contamination of surface water by discharge of partially treated wastewater due to pump station failure.
 - d. Potential for contamination of surface water by runoff of irrigation water from irrigation areas.
14. The Board has reviewed the Negative Declaration and compliance with these waste discharge requirements will mitigate or avoid the significant impacts on air and water quality listed in Finding No. 13. Specific mitigation measures correspond to the alphabetically listed impacts in Finding No. 13 and are as follows:
- a. The Monticello Cemetery will only be irrigated when the public will not have access. The sprinkler systems at the existing spray irrigation field and the Monticello Cemetery should be evaluated for fine mist that has a greater potential for drift. If necessary, the sprinkler heads and distribution system should be modified. Discharge Specification B.15 requires that direct and wind blown spray be confined to the reclamation areas. Specification B.16 prohibits spray irrigation when wind speed exceeds 30 mph.
 - b. Design consideration for odor control should include locating the treatment facility on a well ventilated site, providing sufficient scouring velocities in pipes, providing sufficient aeration in the treatment process for oxidation of the organic compounds, and providing a means to add aeration, chlorine or other disinfectants and odorizing agents to pump stations and manholes. Operational considerations should include providing high pressure connections for hoses for daily flushing of walls and corners to prevent any accumulation of slime or sludge, providing a section in the operations and maintenance manual on odor control, providing regular inspection and maintenance and minimizing the time of draining the last foot or two of water in the storage reservoir. Discharge Specification B.3 prohibits objectionable odors outside the facility.
 - c. A new pump station will be constructed with an automatic switch over to auxiliary power should a power outage occur.

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- d. The existing spray irrigation area will be upgraded for an improved tailwater return system and Monticello Cemetery will be equipped with a tailwater return system. These facilities will eliminate leakage and discharges. Discharge Specification B.17 requires that the wastewater remain in the designated disposal areas at all times.
15. This discharge is exempt from the requirements of Title 23, CCR, Section 2510, et seq. (hereafter Chapter 15). The exemption, pursuant to Section 2511(b), is based on the following:
 - a. The Board is issuing waste discharge requirements, and
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to 22 CCR, Division 4, Chapter 30, as a hazardous waste.
16. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
17. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Spanish Flat Water District, Monticello Cemetery District, and Napa County, their agents, successors and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.

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B. Discharge Specifications:

1. The average monthly dry weather discharge flow shall not exceed 0.025 mgd.
2. The maximum daily discharge shall not exceed 0.053 mgd.
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
4. As a means of discerning compliance with Discharge Specification No. 3, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
5. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
6. The effluent from the treatment facility shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	30	60
Suspended Solids	mg/l	30	60
Settleable Solids	ml/l	0.2	0.5
Coliform	MPN/100ml	23 Median	240

¹ Five-day, 20°C biochemical oxygen demand

7. Treatment plant effluent shall not have a pH less than 6.5 or greater than 8.5.
8. There shall be no standing water in the disposal areas 48 hours after wastewater is applied.
9. Ponds shall be managed to prevent breeding of mosquitos. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

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- b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.**
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.**
- 10. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.**
- 11. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than one foot (measured vertically).**
- 12. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification No. 11.**
- 13. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitos. More specifically,**
 - a. Tail water must be returned and all applied irrigation water must infiltrate completely within a 48-hour period.**
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.**
 - c. Low-pressure and unpressurized pipelines and ditches accesssible to mosquitos shall not be used to store reclaimed water.**
- 14. Reclaimed water for irrigation shall be managed to minimize erosion, runoff, and movement of aerosols from the disposal areas.**
- 15. Direct or windblown spray shall be confined to the designated reclamation areas and prevented from contacting drinking water facilities.**
- 16. The Discharger may not spray irrigate effluent during periods of precipitation and for at least 24 hours after cessation of precipitation, or when winds exceed 30 mph.**

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17. The discharge shall remain within designated disposal areas at all times.
18. Signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal to alert the public of the use of reclaimed water.
19. Stormwater runoff from the irrigation areas shall not be discharged to any surface water drainage course within 48-hours of the last application of reclaimed water.
20. Application of reclaimed wastewater to the reclamation area shall be at reasonable rates considering the crop, soil, climate and irrigation management system. The nutrient loading of the reclamation area, including the nutritive value of organic and chemical fertilizers and of the reclaimed water, shall not exceed the crop demand.
21. There shall be no irrigation or impoundment of reclaimed water within 500 feet of any domestic water well or within 100 feet of any irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that less distance is justified.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.
3. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled *Manual of Good Practice for Landspreading of Sewage Sludge*, and *Standards for the Use or Disposal of Sewage Sludge*, issued by the U.S. Environmental Protection Agency in 40 CFR 503.

D. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to:

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1. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified.
2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.
3. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
4. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.
5. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
6. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

E. Provisions:

1. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, ground water, cooling waters, and condensates that are essentially free of pollutants.
2. The Discharger shall use the best practical cost-effective control technique currently available to limit mineralization to no more than a reasonable increment.
3. The Discharger shall implement the necessary legal authorities, programs, and controls to ensure that the following incompatible wastes are not introduced to the treatment system, where incompatible wastes are:
 - a. Wastes which create a fire or explosion hazard in the treatment works;
 - b. Wastes which will cause corrosive structural damage to treatment works, but in no case wastes with a pH lower than 5.0, unless the works is specially designed to accommodate such wastes;

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- c. Solid or viscous wastes in amounts which cause obstruction to flow in sewers, or which cause other interference with proper operation or treatment works;
 - d. Any waste, including oxygen demanding pollutants (BOD, etc.), released in such volume or strength as to cause inhibition or disruption in the treatment works, and subsequent treatment process upset and loss of treatment efficiency;
 - e. Heat in amounts that inhibit or disrupt biological activity in the treatment works, or that raise influent temperatures above 40°C (104°F), unless the treatment works is designed to accommodate such heat;
 - f. Petroleum oil, nonbiodegradable cutting oil, or products of mineral oil origin in amounts that will cause interference or pass through;
 - g. Pollutants which result in the presence of toxic gases, vapors, or fumes within the treatment works in a quantity that may cause acute worker health and safety problems; and
 - h. Any trucked or hauled pollutants, except at points predesignated by the Discharger.
- 4. For the purposes of this Order, "spray irrigation" means application of reclaimed water to crops by sprinklers.
 - 5. Reclaimed water controllers, valves, etc., shall be affixed with reclaimed water warning signs, and these and quick couplers and sprinkler heads shall be of a type, or secured in a manner, that permits operation by authorized personnel only.
 - 6. Piping, valves and outlets shall be marked to differentiate reclaimed water from domestic water supply.
 - 7. Supplementing reclaimed water by connection with a domestic drinking water source or irrigation or industrial wells requires an air gap device.
 - 8. A contingency plan, including notification of the Board and health agencies and outlining action to be taken when effluent quality fails to meet required standards, shall be submitted within 60 days of adoption of this Order.
 - 9. Irrigation shall be scheduled for times the public is not present. Therefore times of irrigation shall be limited to between the hours of 9:00 p.m. and 5:00 a.m.

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10. Potable water should be supplied for flower containers to avoid public use of reclaimed water that could result in splashing or spray or contaminated hands.
11. If the Discharger intends to use reclaimed water on crops other than those specified in an accepted land management plan, it shall first submit a written report demonstrating, to the satisfaction of the Executive Officer, that management of reclaimed water and irrigated properties will assure compliance with the terms of this Order.
12. The Discharger shall comply with the Monitoring and Reporting Program No. 93-236 which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
13. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
14. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
15. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
16. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
17. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

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18. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.
19. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 3 December 1993.



WILLIAM H. CROOKS, Executive Officer

KMG:ldj

Attachments



California Regional Water Quality Control Board Central Valley Region

Robert Schneider, Chair



Arnold
Schwarzenegger
Governor

Linda S. Adams
Secretary for
Environmental
Protection

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29 September 2006

CERTIFIED MAIL

7005 1160 0005 0478 4118

Paul Roussopoulos
Spanish Flat Water District
4340 Spanish Flat Loop Road
P.O. Box 9163
Napa, CA 94559

CERTIFIED MAIL

7005 1160 0005 0478 4125

Monticello Public Cemetery District
c/o Martha Burdick
Napa County Dept. of Public Works
1195 Third Street, Room 201
Napa, CA 94559

CERTIFIED MAIL

7005 1160 0005 0478 4132

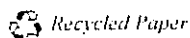
Napa County
Board of Supervisors
1195 Third Street, Suite 310
Napa, CA 94559

NOTICE OF ADOPTION
OF
THIRD REVISION OF MONITORING AND REPORTING PROGRAM NO. 93-236
FOR
SPANISH FLAT WATER DISTRICT
MONTICELLO CEMETARY DISTRICT
AND NAPA COUNTY
SPANISH FLAT WASTEWATER TREATMENT AND DISPOSAL FACILITY
NAPA COUNTY

The California Regional Water Quality Control Board, Central Valley Region, adopted Order No. R5-2006-0095 for the Third Revision of Monitoring and Reporting Program No. 93-236 at its 22 September 2006 meeting.

Please note that the Monitoring and Reporting Program (MRP) contains compliance schedules and specific timetables for submitting reports and conducting studies of the wastewater system. The first scheduled due date is 1 November 2006, when you must submit the name of the California Registered Professional that will prepare the Groundwater Monitoring Well Installation Workplan (due by 1 March 2007) and the Well Installation Report (due by 1 October 2007). The MRP also contains specified monitoring requirements for you to implement. Please review the MRP closely so that you may establish the appropriate sampling schedules and protocols. A copy of the Order must be maintained at the facility with existing Waste Discharge Requirements Order No. 93-236 (copy enclosed), and must be accessible to anyone operating the waste treatment system.

California Environmental Protection Agency



In order to conserve paper and reduce mailing costs, a paper copy of the order has been sent only to the Discharger. Interested parties are advised that the full text of this order is available on the Regional Board's web site at

http://www.waterboards.ca.gov/centralvalley/adopted_orders/index.html.

Anyone without access to the Internet who needs a paper copy of the order can obtain one by calling Regional Water Board staff.

If you have any questions, please contact Guy Childs at (916) 464-4648.


MARK R. LIST, P.G. Chief
Waste Discharge to Land Unit

Enc: MRP Order No. R5-2006-0095
WDRs Order No. 93-236

cc w/ enc: Wesley Plunkett, Spanish Flat Water District, Napa
Richard Thomasser, Napa County Public Works, Napa
Roberta L. Larson, Somach, Simmons & Dunn, Sacramento

ORDER NO. R5-2006-0095

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

THIRD REVISION OF MONITORING AND REPORTING PROGRAM NO. 93-236

FOR
SPANISH FLAT WATER DISTRICT
MONTICELLO CEMETERY DISTRICT
AND NAPA COUNTY
SPANISH FLAT WASTEWATER TREATMENT AND DISPOSAL FACILITY
NAPA COUNTY

This Monitoring and Reporting Program (MRP) rescinds and replaces the Second Revision of MRP No. 93-236. The MRP incorporates requirements for monitoring of the wastewater treatment facility. This MRP is issued pursuant to Water Code Section 13267. The Staff Report (attached) provides the report requirements in this MRP. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Regional Board or the Executive Officer.

All wastewater samples should be representative of the volume and nature of the discharge. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form. Process wastewater flow monitoring shall be conducted continuously using a flow meter and shall be reported in cumulative gallons per day.

Field test instruments (such as pH and dissolved oxygen) may be used provided that:

1. The operator is trained in the proper use of the instrument;
2. The instruments are field calibrated prior to each monitoring event;
3. Instruments are serviced and/or calibrated per manufacturer's recommended frequency; and
4. Field calibration reports are submitted as described in the "Reporting" section of this MRP.

INFLUENT MONITORING

Influent samples shall be collected just prior to the package treatment plant. Influent monitoring shall consist of the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gpd	Continuous Meter	Daily	Monthly
BOD ₅ ¹	gpd	Grab	Monthly	Monthly

¹ 5-day, 20°C Biochemical Oxygen Demand.

EFFLUENT MONITORING

Effluent samples shall be collected from the pipeline that discharges to the evaporation/percolation pond. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Total Coliform Organisms ¹	MPN/100 mL ²	Grab	Weekly	Monthly
BOD ₅ ³	mg/L	Grab	Twice Monthly	Monthly
Settleable Solids	mg/L	Grab	Monthly	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly	Monthly
Nitrate as Nitrogen	mg/L	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/L	Grab	Monthly	Monthly
Standard Minerals ⁴	mg/L	Grab	Annually	Annually

¹ Using a minimum of 10 tubes or two dilutions.

² Most probable number per 100 ml.

³ 5-day, 20°C Biochemical Oxygen Demand.

⁴ Standard Minerals shall include the following: boron, calcium, iron, manganese, magnesium, potassium, sodium, chloride, total alkalinity (including alkalinity series), and hardness.

POND MONITORING

The percolation/evaporation pond shall be monitored for the parameters specified below. Freeboard shall be measured vertically from the surface of the pond water to the lowest elevation of the surrounding levee and shall be measured to the nearest 0.1 feet.

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Dissolved Oxygen ^{1,4}	mg/L	Grab	Weekly	Monthly
pH ⁴	pH units	Grab	Weekly	Monthly
Freeboard	0.1 feet	Observation	Weekly	Monthly
Berm Seepage ²	NA	Observation	Weekly	Monthly
Odors ³	--	Observation	Weekly	Monthly

¹ Samples shall be collected at a depth of one foot from each pond in use, opposite the inlet. Samples shall be collected between 0700 and 0900 hours.

² Reservoir containment levees shall be observed for signs of seepage or surfacing water along the exterior toe of the levees and dam. If surfacing water is found, then a sample shall be collected and tested for total coliform organisms and total dissolved solids.

³ The presence of strong or unusual odors shall be reported.

⁴ Hand held meter may be used.

DISPOSAL AREA MONITORING

The following Disposal Area Monitoring shall be required only if wastewater has been delivered to either area during the monthly reporting period. If no disposal has taken place that month, the monitoring report shall so state.

Monitoring of the two disposal areas shall be conducted **daily** and the results shall be included in the monthly monitoring report. Evidence of erosion, saturation, irrigation runoff, or the presence of nuisance conditions shall be noted in the report. Effluent monitoring results shall be used in calculations to ascertain loading rates at the application area. Monitoring of the disposal area shall include the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	Gallons	Meter	Daily	Monthly
Local Rainfall	Inches	Measurement	Daily	Monthly
Acreage Applied ¹	Acres	Calculated	Daily	Monthly
Application Rate ²	gal/acre/day	Calculated	Daily	Monthly
BOD ₅ Loading Rate ²	lbs/acre/day ³	Calculated	Monthly	Monthly
Total Nitrogen Loading Rate ²	lbs/acre/month ³	Calculated	Monthly	Monthly
Total Dissolved Solids Loading Rate ²	lbs/acre/month ³	Calculated	Monthly	Monthly

¹ Land application areas shall be identified.

² For each land application area.

³ Report monthly total and cumulative annual to date.

The entire irrigated areas shall be inspected weekly during or immediately following an irrigation event to identify any equipment malfunction or other circumstances that might allow irrigation runoff to leave the irrigation area and/or create ponding conditions that violate the Waste Discharge Requirements. A log of these inspections shall be kept at the facility and made available for review upon request.

SLUDGE MONITORING

In accordance with EPA's POTW Sludge Sampling and Analysis Guidance Document, August 1989, a composite sample of sludge shall be collected when removed from the evaporation/percolation pond and tested for the following metals:

Cadmium	Copper	Nickel
Chromium	Lead	Zinc

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

GROUNDWATER MONITORING

By **1 November 2006**, the Discharger shall submit the name of the California Registered Professional that will prepare the two reports listed below.

By **1 March 2007**, the Discharger shall submit a Groundwater Monitoring Well Installation Workplan prepared in accordance with, and including the items listed in, the first section of the attached "*Items to be Included in a Monitoring Well Installation Workplan and Monitoring Well Installation Report of Results.*" The workplan shall describe a groundwater monitoring network consisting of at least three wells around the storage pond specifically designed to ensure that background groundwater quality is adequately characterized and any potential water quality impacts from the discharge are detected. One of the three wells may consist of the County Maintenance Yard well, if the workplan provides details showing that the well is adequate to monitor the storage pond or background groundwater quality. The system shall be designed to yield samples representative of the uppermost portion of the first aquifer underlying the facility site. The workplan shall also include a Sampling and Analysis Plan (SAP) that includes all information listed in the second section of the attachment to this MRP.

By **1 October 2007**, the Discharger shall submit a Well Installation Report that presents well construction, well development, and well surveying details, and as well as any soil sampling details, and contains the information listed in the second section of the attachment to this MRP.

The groundwater monitoring program shall begin in the **fourth quarter of 2007**. Samples shall be collected from all groundwater monitoring wells at the facility. Prior to construction of any new groundwater monitoring wells, the Discharger shall submit plans and specifications to the Regional Board for review and approval. Once installed, all new wells shall be added to the MRP and shall be sampled and analyzed according to the schedule below.

Prior to sampling, the groundwater elevations shall be measured and the wells shall be purged at least three well volumes until temperature, pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Samples shall be collected using standard EPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency³</u>
Depth to Groundwater	0.01 feet	Measurement	Quarterly	Quarterly
Groundwater Elevation ¹	0.01 feet	Calculated	Quarterly	Quarterly

Gradient	feet/feet	Calculated	Quarterly	Quarterly
Gradient Direction	Degrees	Calculated	Quarterly	Quarterly
Total Coliform Organisms	MPN/100 mL	Grab	Quarterly	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly	Quarterly
Nitrate as Nitrogen	mg/L	Grab	Quarterly	Quarterly
Total Kjeldahl Nitrogen	mg/L	Grab	Quarterly	Quarterly
Standard Minerals ²	mg/L	Grab	Annually	Annually

¹ Groundwater elevation shall be determined based on depth-to-water measurements from a surveyed measuring point elevation on the well.

² Standard Minerals shall include the following: boron, calcium, iron, manganese, magnesium, potassium, sodium, chloride, total alkalinity (including alkalinity series), and hardness.

³ Beginning with the fourth quarter 2007

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., process wastewater effluent, groundwater well, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Quarterly Groundwater Monitoring Reports and the annual groundwater evaluation shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

Section 13267 of the California Water Code state in part: (b)(1) *In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports.*

Section 13268 of the California Water Code states in part: (a) *Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of Section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of Section 13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).*

(b)(1) Civil liability may be administratively imposed by a regional board in accordance with Article 2.5 (commencing with Section 13323) of Chapter 5 for a violation of subdivision (a) in an amount which shall not exceed one thousand dollars (\$1,000) for each day in which the violation occurs.

The Discharger owns and operates the facility that discharges waste subject to WDRs Order No. 93-236. The following reports are required to ensure compliance with the WDRs and the Revised MRP. Pursuant to Section 13267 of the California Water Code, the Discharger shall submit the following reports (as well as the above groundwater monitoring workplan and installation report) by the specified due dates:

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board by the **1st day of the second month** following the end of the reporting period (i.e. the August monthly report is due by 1 October). The monthly reports shall include the following:

1. Results of influent, effluent, pond, and disposal area monitoring;
2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format;
3. If requested by staff, copies of laboratory analytical report(s); and
4. A calibration log verifying calibration of all hand held monitoring instruments and devices used to comply with the prescribed monitoring program;

B. Quarterly Monitoring Reports

Beginning with the **fourth quarter of 2007**, the Discharger shall establish a quarterly sampling schedule for groundwater monitoring such that samples are obtained approximately every three months. Quarterly monitoring reports shall be submitted to the Regional Board by the **1st day of the second month after the quarter** (i.e. the January-March quarterly report is due by May 1st) and may be combined with the monthly report. The Quarterly Report shall include the following:

1. Results of groundwater monitoring;
2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities. The narrative shall be sufficiently detailed to verify compliance with the WDRs, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of the casing volume; and total volume of water purged;

3. Calculation of groundwater elevations, an assessment of the groundwater flow direction and gradient on the date of measurement, comparison to previous flow direction and gradient data, and discussion of seasonal trends, if any;
4. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable);
5. A comparison of monitoring data to the discharge specifications, groundwater limitations, and surface water limitations, and explanation of any violation of those requirements;
6. Summary data tables of historical and current water table elevations and analytical results;
7. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and other sampling stations, and groundwater elevation contours referenced to mean sea level datum; and
8. Copies of laboratory analytical report(s) for groundwater monitoring.

C. Annual Monitoring Reports

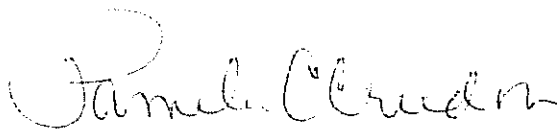
An Annual Report shall be prepared as the fourth quarter monitoring report and shall include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by **1 February of each year** and shall include the following:

1. The contents of the regular groundwater monitoring report for the last sampling event of the year;
2. If requested by staff, tabular and graphical summaries of all data collected during the year;
3. Data for the effluent and groundwater monitoring performed on an annual basis;
4. An evaluation of the groundwater quality beneath the facility;
5. A discussion of whether sludge was removed from the pond, and if so, the results of the sampling;
6. An evaluation of the performance of the wastewater treatment system, as well as a forecast of the flows anticipated in the next year;
7. Verification of appropriate employee training for all personnel involved in operation and maintenance of wastewater treatment system;

8. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements; and
9. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

A letter transmitting the self-monitoring reports shall accompany each report. Such a letter shall include a discussion of requirement violations found during the reporting period, and actions taken or planned for correcting noted violations, such as operation or facility modifications. If the discharger has previously submitted a report describing corrective actions and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. The transmittal letter shall contain a statement by the discharger, or the discharger's authorized agent, under penalty of perjury, that to the best of the signer's knowledge the report is true, accurate and complete.

I, PAMELA C. CREEDON, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 22 September 2006.



PAMELA C. CREEDON, Executive Officer

Attachment: Items to be Included in a Monitoring Well Installation Workplan and Monitoring Well
Installation Report of Results
2006 Staff Report



California Regional Water Quality Control Board Central Valley Region

Robert Schneider, Chair

Linda S. Adams
Secretary for
Environmental
Protection

Sacramento Main Office
11020 Sun Center Drive #200, Rancho Cordova, California 95670-6114
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<http://www.waterboards.ca.gov/centralvalley>



Arnold
Schwarzenegger
Governor

MRP ORDER NO. R5-2006-0095 ATTACHMENT A REQUIREMENTS FOR MONITORING WELL INSTALLATION WORKPLANS AND MONITORING WELL INSTALLATION REPORTS

Prior to installation of groundwater monitoring wells, the Discharger shall submit a workplan containing, at a minimum, the information listed in Section 1, below. Wells may be installed after staff approve the workplan. Upon installation of the monitoring wells, the Discharger shall submit a well installation report which includes the information contained in Section 2, below. All workplans and reports must be prepared under the direction of, and signed by, a registered geologist or civil engineer licensed by the State of California.

SECTION 1 - Monitoring Well Installation Workplan and Groundwater Sampling and Analysis Plan

The monitoring well installation workplan shall contain the following minimum information:

A. General Information:

- Purpose of the well installation project
- Brief description of local geologic and hydrogeologic conditions
- Proposed monitoring well locations and rationale for well locations
- Topographic map showing facility location, roads, and surface water bodies
- Large scaled site map showing all existing on-site wells, proposed wells, surface drainage courses, surface water bodies, buildings, waste handling facilities, utilities, and major physical and man-made features

B. Drilling Details:

- On-site supervision of drilling and well installation activities
- Description of drilling equipment and techniques
- Equipment decontamination procedures
- Soil sampling intervals (if appropriate) and logging methods

C. Monitoring Well Design (in narrative and/or graphic form):

- Diagram of proposed well construction details
 - Borehole diameter
 - Casing and screen material, diameter, and centralizer spacing (if needed)
 - Type of well caps (bottom cap either screw on or secured with stainless steel screws)
 - Anticipated depth of well, length of well casing, and length and position of perforated interval
 - Thickness, position and composition of surface seal, sanitary seal, and sand pack

California Environmental Protection Agency

- Anticipated screen slot size and filter pack

D. Well Development (not to be performed until at least 48 hours after sanitary seal placement):

Method of development to be used (i.e., surge, bail, pump, etc.)

Parameters to be monitored during development and record keeping technique

Method of determining when development is complete

Disposal of development water

E. Well Survey (precision of vertical survey data shall be at least 0.01 foot):

Identify the Licensed Land Surveyor or Civil Engineer that will perform the survey

Datum for survey measurements

List well features to be surveyed (i.e. top of casing, horizontal and vertical coordinates, etc.)

F. Schedule for Completion of Work

G. **Appendix: Groundwater Sampling and Analysis Plan (SAP)**

The Groundwater SAP shall be included as an appendix to the workplan, and shall be utilized as a guidance document that is referred to by individuals responsible for conducting groundwater monitoring and sampling activities.

Provide a detailed written description of standard operating procedures for the following:

- Equipment to be used during sampling
- Equipment decontamination procedures
- Water level measurement procedures
- Well purging (include a discussion of procedures to follow if three casing volumes cannot be purged)
- Monitoring and record keeping during water level measurement and well purging (include copies of record keeping logs to be used)
- Purge water disposal
- Analytical methods and required reporting limits
- Sample containers and preservatives
- Sampling
 - General sampling techniques
 - Record keeping during sampling (include copies of record keeping logs to be used)
 - QA/QC samples
- Chain of Custody
- Sample handling and transport

SECTION 2 - Monitoring Well Installation Report

The monitoring well installation report must provide the information listed below. In addition, the report must also clearly identify, describe, and justify any deviations from the approved workplan.

A. General Information:

Purpose of the well installation project

Brief description of local geologic and hydrogeologic conditions encountered during installation of the wells

Number of monitoring wells installed and copies of County Well Construction Permits

Topographic map showing facility location, roads, surface water bodies

Scaled site map showing all previously existing wells, newly installed wells, surface water bodies, buildings, waste handling facilities, utilities, and other major physical and man-made features.

B. Drilling Details (in narrative and/or graphic form):

On-site supervision of drilling and well installation activities

Drilling contractor and driller's name

Description of drilling equipment and techniques

Equipment decontamination procedures

Soil sampling intervals and logging methods

Well boring log

- Well boring number and date drilled
- Borehole diameter and total depth
- Total depth of open hole (same as total depth drilled if no caving or back-grouting occurs)
- Depth to first encountered groundwater and stabilized groundwater depth
- Detailed description of soils encountered, using the Unified Soil Classification System

C. Well Construction Details (in narrative and/or graphic form):

Well construction diagram, including:

- Monitoring well number and date constructed
- Casing and screen material, diameter, and centralizer spacing (if needed)
- Length of well casing, and length and position of perforated interval
- Thickness, position and composition of surface seal, sanitary seal, and sand pack
- Type of well caps (bottom cap either screw on or secured with stainless steel screws)

E. Well Development:

Date(s) and method of development

How well development completion was determined

Volume of water purged from well and method of development water disposal

Field notes from well development should be included in report

F. Well Survey (survey the top rim of the well casing with the cap removed):

Identify the coordinate system and datum for survey measurements

Describe the measuring points (i.e. ground surface, top of casing, etc.)

Present the well survey report data in a table

Include the Registered Engineer or Licensed Surveyor's report and field notes in appendix

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 93-236

FOR
SPANISH FLAT WATER DISTRICT
MONTICELLO CEMETERY DISTRICT
AND NAPA COUNTY
SPANISH FLAT WASTEWATER TREATMENT AND DISPOSAL FACILITY
NAPA COUNTY

Specific sample station locations shall be established under direction of the Board's staff and a description of the stations shall be attached to this Order.

EFFLUENT MONITORING

Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
20°C BOD ₅	mg/l	Grab	Monthly
Suspended Matter	mg/l	Grab	Weekly
Settleable Matter	ml/l	Grab	Weekly
Nitrates as N	mg/l	Grab	Monthly
Total Dissolved Solids	mg/l	Grab	Quarterly
Specific Conductivity	μmhos/cm	Grab	Quarterly
Standard Minerals	mg/l	Grab	Annually
pH	pH Units	Grab	Weekly
Total Coliform Organisms	MPN/100 ml	Grab	Weekly
Flow	mgd	Cumulative	Daily
Oil & Grease	mg/l	Grab	Quarterly

SLUDGE MONITORING

A composite sample of sludge shall be collected annually in accordance with EPA's POTW Sludge Sampling and Analysis Guidance Document, August 1989, and tested for the following metals:

Arsenic	Mercury
Cadmium	Molybdenum
Chromium	Nickel
Copper	Selenium
Lead	Zinc

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the municipal water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
Electrical Conductivity @ 25°C	μmhos/cm	Annual

ANNUAL LAND MANAGEMENT REPORTS

The Discharger shall submit a summary report on the land management operation after the conclusion of each season. The report shall discuss total water application over the season; the total volume of wastewater applied; the total nutrient loading from wastewater, sludges, and chemical fertilizers; and amount of nutrients removed through harvest of the crop. In short, the report shall present a mass balance relative to pollutants of concern and hydraulic loading. The report is due by 30 January of the following year.

MONITORING AND REPORTING PROGRAM
SPANISH FLAT WATER DISTRICT, ET AL.
SPANISH FLAT WASTEWATER TREATMENT AND DISPOSAL FACILITY
NAPA COUNTY

-3-

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements.

Monthly monitoring reports shall be submitted to the Regional Board by the **20th day** of the following month.

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

Upon written request of the Board, the Discharger shall submit a report to the Board by **30 January** of each year. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the Discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by: William H. Crooks
WILLIAM H. CROOKS, Executive Officer

3 December 1993

(Date)

9/22/93:KMG:kmg/ldj

INFORMATION SHEET

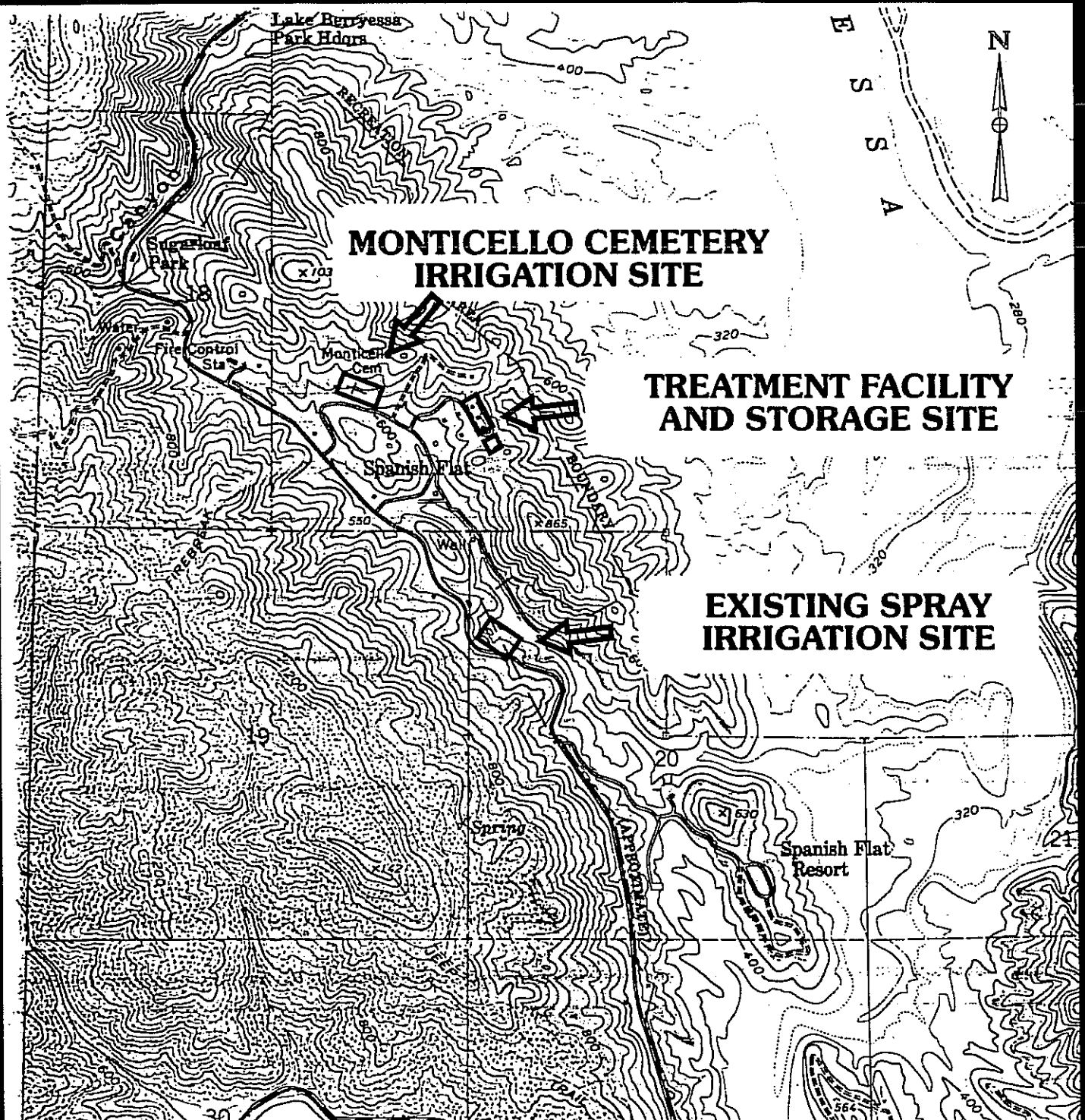
**SPANISH FLAT WATER DISTRICT,
MONTICELLO CEMETERY DISTRICT
AND NAPA COUNTY
SPANISH FLAT WASTEWATER TREATMENT AND DISPOSAL FACILITY
NAPA COUNTY**

This wastewater treatment facility is being built to replace the existing wastewater treatment facility which has inadequate capacity and also a history of seepage problems from its evaporation/percolation ponds which resulted in the issuance of Cleanup and Abatement Order No. 89-715. The facility will provide sewerage service to Spanish Flat Mobile Villa, Spanish Flat Woodlands and Spanish Flat Center.

The treatment system is an extended aeration package treatment plant with an aeration tank, a clarifier, and a chlorine contact chamber. Wastewater will be discharged to a holding pond with eventual disposal to two spray irrigation areas. One of these areas is the existing spray irrigation field, the other is Monticello Cemetery. The irrigation area at Monticello Cemetery will have a moderate degree of public contact. The secondary treatment process will meet the effluent standards required by Title 22 for reclamation purposes.

The facility will have a maximum daily discharge of 0.053 mgd and an average monthly discharge of 0.025 mgd. Surface water in the vicinity of the facility is limited to ephemeral streams which are tributaries to Lake Berryessa.

12/6/93:KMG:ldj



ATTACHMENT A

**SPANISH FLAT WATER DISTRICT, ET AL.
SPANISH FLAT WASTEWATER TREATMENT AND DISPOSAL FACILITY
NAPA COUNTY**

**U.S.G.S. LAKE BERRYESSA 7.5 QUAD
SEC. 17, 18, & 20, T8N, R3W, MDB&M
SCALE 1 Inch = 2000 feet**